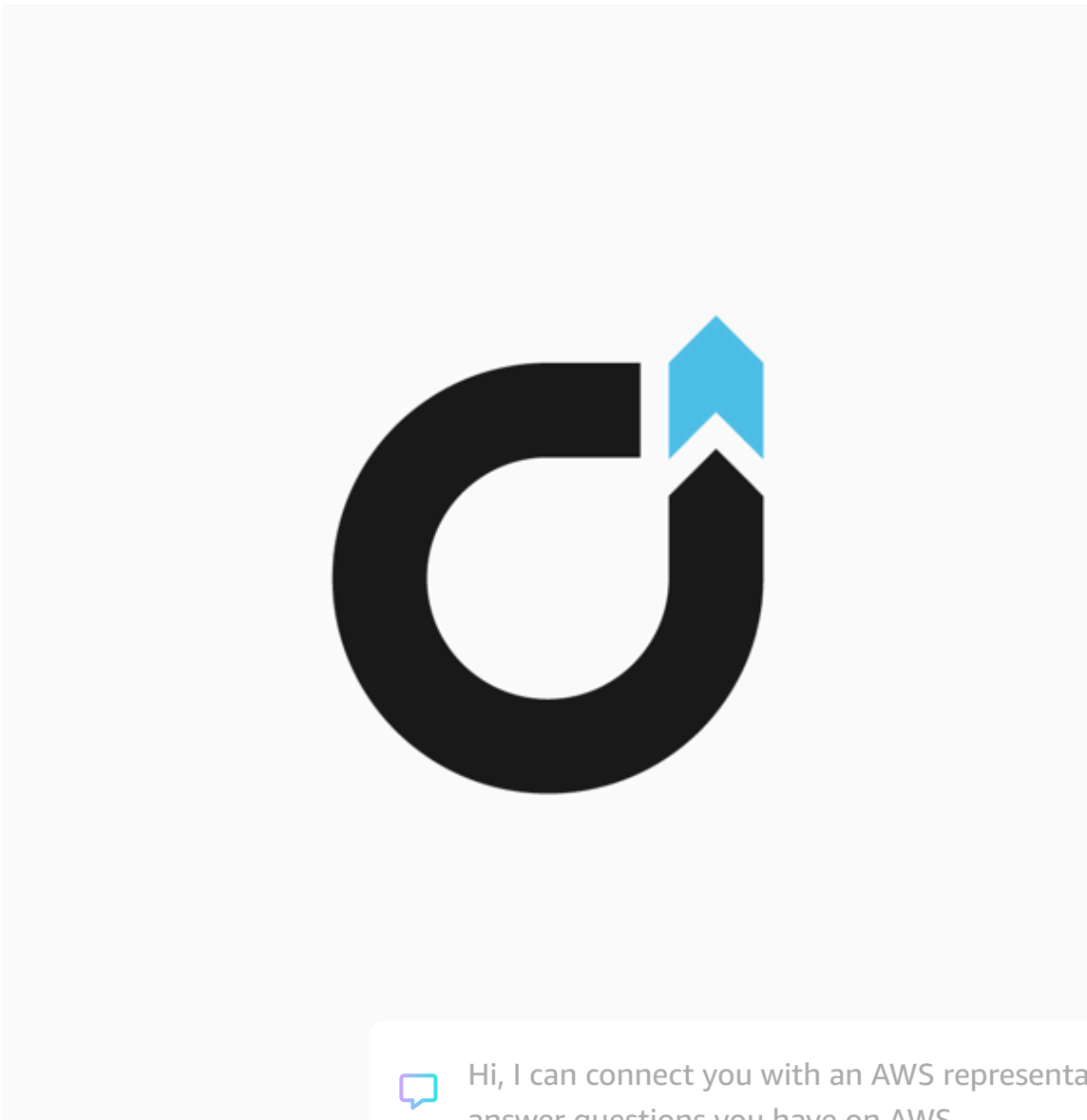


2023



Deploying AI/ML a Sustainable Smart

Learn how Omniflow helps its customers reduce carbon emissions with their sustainable smart lampposts using AWS and NVIDIA.



Hi, I can connect you with an AWS representative or answer questions you have on AWS.



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



2-4 cameras supported

per pole

Uses existing

light poles

Keeps costs low

for customers

Promotes sustainable

cities

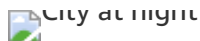
Uses renewable energy

sources

Overview

Imagine a world where a lamppost does more than just illuminate streets; it actively contributes to a smarter, safer, and more sustainable community. Using Amazon Web Services (AWS) and NVIDIA technologies, [Omniflow](#) is turning this vision into a reality.

Omniflow's smart lampposts use artificial intelligence and machine learning (AI/ML) algorithms and computer vision to help communities become more sustainable and energy efficient. By adopting edge computing and a cloud-based technology stack, the company has achieved significant operational benefits—and is helping its customers reduce carbon emissions with lighting powered by wind and solar patented technology.



Opportunity | Using AWS and NVIDIA Services to Power Advanced AI/ML Use Cases for Sustainability with Omniflow

Founded in 2012, Omniflow provides smart-city solutions to government entities as well as private businesses. Traditional lampposts require a significant amount of energy to operate and can be costly to maintain. To reduce grid power consumption, Omniflow retrofits old lights with its smart lamppost technology. This solution is not only more sustainable than traditional options but also transforms lampposts into valuable sources of data.

“Our smart lampposts are powered by wind and solar at the edge and completely integrated into a single infrastructure,” says Pedro Ruão, founder and CEO of Omniflow. “We use these assets to provide many other services, such as air-quality monitoring, computer vision, and traffic analytics.”

With Omniflow's lampposts, its customers can track various data points related to their environment and make decisions to improve efficiency, sustainability, and public safety. As the company started to incorporate more features, such as cameras and sensors, it needed more computational power to process data and run its AI/ML algorithms. To resolve this challenge, the company joined [NVIDIA Inception](#), a free program to help companies solve faster through cutting-edge technology.



“During NVIDIA Inception, we were able to purchase some initial hardware for testing,” says Ruão. “We liked the results, so we started working with the NVIDIA and AWS teams more closely.” To dive deeper into AWS services, Omniflow also joined the [AWS Sustainable Cities Accelerator](#) program—an initiative designed to help mature startups accelerate their impact, access additional resources, and expand their reach.

Solution | Adopting Amazon EC2 P4 Instances and NVIDIA Jetson Nano to Process Computer Vision Data at the Edge

Omniflow migrated to a new system built on [Amazon Elastic Compute Cloud](#) (Amazon EC2) [P4 Instances](#). These instances are powered by NVIDIA A100 Tensor Core GPUs and deliver industry-leading high throughput and low-latency networking for Generative AI workloads. Omniflow is currently using Amazon EC2 P4 Instances with [NVIDIA TAO Toolkit](#), a service that helps quickly train and customize object detection models powered by AI/ML at the edge.

Within each smart lamppost, Omniflow has embedded [NVIDIA Jetson Nano](#), which brings accelerated AI performance to the edge. After an AI/ML model is trained, it is deployed to NVIDIA Jetson Nano devices in each lamppost. The models are then used to perform different functions, such as analyzing traffic footage or assessing pollution levels. “We were already performing edge computing, but computer vision requires a much more powerful standard of performance,” says Ruão. “With NVIDIA Jetson, each of our lampposts has the computing power to process two to four cameras and extract all the necessary metrics, plus perform air-quality monitoring.”

To send this data back to AWS, Omniflow uses [AWS IoT Core](#), a service that helps developers connect billions of Internet of Things (IoT) devices and route trillions of messages to AWS services without managing infrastructure. This information is processed using functions running on [AWS Lambda](#)—a serverless, event-driven compute service—and stored in Omniflow’s database. These AWS Lambda functions activate specific actions based on the incoming data, such as abnormal air-quality levels or unusual traffic patterns.

For instance, if the air-quality sensors detect elevated levels of pollutants, an AWS Lambda function activates an alert to municipal authorities. Similarly, if the computer vision models identify a traffic jam or an accident, the system automatically alerts traffic management centers for immediate action. “By gathering data on AWS, we can help our customers optimize their environments,” says Ruão.

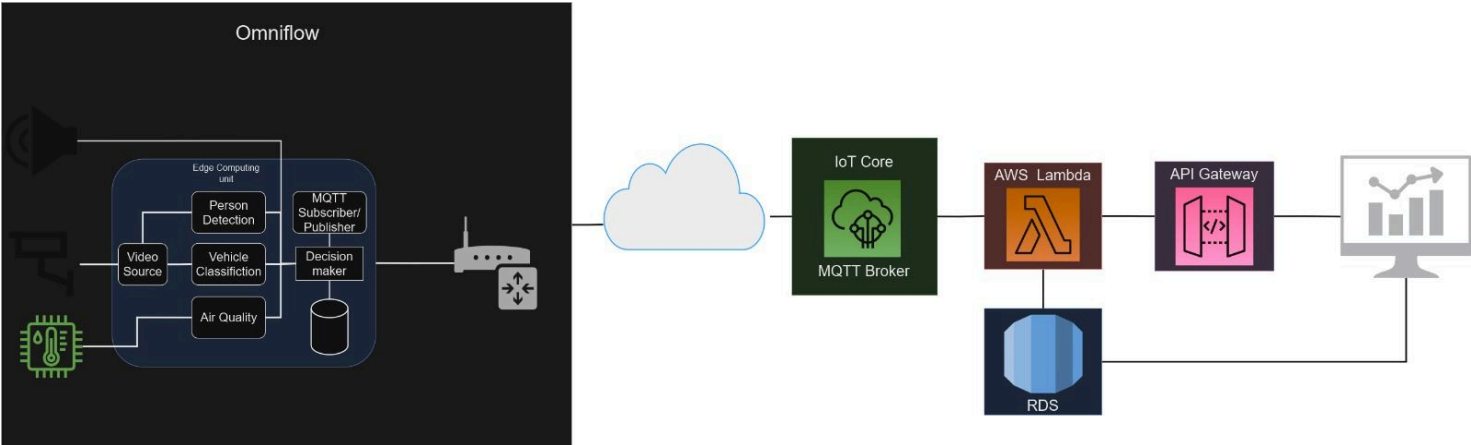


Although the lampposts are designed to run continuously, they don't require constant energy. They primarily use renewable energy sources like wind and solar and are connected to the existing electrical grid for backup. By taking this approach, Omniflow claims that it is helping to reduce the lampposts' carbon emissions by over 90 percent. The lampposts are also cost effective; because Omniflow's system is multipurpose and integrated, customers do not need to build or maintain separate infrastructure for each service.

With this highly modular and serverless architecture, Omniflow can scale its services and adapt to the varied requirements of different customers. Moreover, because the entire system is cloud based, the company can push updates and improvements to edge devices quickly so that its smart lampposts are equipped with the latest features and capabilities.

Omniflow also deploys faster on the cloud. "One of the main benefits of using AWS is the speed of development," says Luis Carvalho, research and development engineer at Omniflow. "Because of the way AWS services are built, it is much faster to build solutions using tools from AWS than building from scratch or using other services that we have tried previously." Omniflow's usual time to market is typically 2 months from order to full deployment; however, on AWS, it completed a major installation in just 15 days for a special event.

Architecture Diagram



Outcome | Helping Communities Become Safer and More Sustainable on the Cloud

Omniflow successfully used AWS and NVIDIA AI and Generative AI technologies to create smart lampposts that contribute to sustainable, energy-efficient communities. With this stack, the company successfully resolved its computational challenges—and achieved environmental and operational benefits along the way. Omniflow plans to expand its services in the coming years, offering cities the flexibility to integrate new functionalities into their existing infrastructure.

"We started as a renewable energy creation company," says Ruão. "But now, by using NVIDIA and AWS technologies, we have unlocked a world of data."

About Omniflow

Founded in 2012, Omniflow is a provider of smart-city solutions. It serves public and private customers around the world.

AWS Services Used

Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

[Learn more »](#)

Amazon SageMaker

Amazon SageMaker is built on Amazon's two decades of experience developing real-world ML applications, including product recommendations, personalization, intelligent shopping, robotics, and voice-assisted devices.

[Learn more »](#)

Amazon SageMaker Studio

Amazon SageMaker Studio provides a single, web-based visual interface where you can perform all ML development steps, improving data science team productivity by up to 10x..

[Learn more »](#)

Amazon SageMaker Pipelines

Amazon SageMaker Pipelines is the first purpose-built, easy-to-use continuous integration and continuous delivery (CI/CD) service for machine learning (ML).

[Learn more »](#)



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Revolutionizing Real Estate Transactions Using AI on...

Rexera revolutionizes real estate by automating the manual, error-prone closing workflows in real estate transactions, minimizing human involvement so that real estate professionals can speed up transactions, reduce friction, and lower costs.

2024

PORTUGAL



Omniflow Deploys AI/ML for Smart Lampposts...

Omniflow's smart lampposts use artificial intelligence and machine learning (AI/ML) algorithms and computer vision to help communities become more sustainable and energy efficient. Imagine a world where a lamppost does more than just illuminate streets; it actively contributes to a smarter, safer, and more sustainable community. Using Amazon



2023

AUSTRALIA



Victorian Land Registry Services Ensures...

Victorian Land Registry Services (VLRs) uses AWS to deliver near 100 percent availability for a critical web application, reduce disaster recovery time from 16 hours to under 30



IRELAND



CADFEM Case Study

CADFEM started using Amazon Web Services (AWS) in 2010 to bolster its internal compute resources when doing consulting work for clients. Soon after, more clients wanted to know about AWS, and CADFEM began sharing Amazon Machine Images (AMIs) with clients who wanted to do the same.

2019



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